Training Statistics Report: FastGS on MipNeRF-360 Counter

- **Scene**: `counter`
- **Dataset Source**: `mipnerf360`
- **Git Branch**: `main`
- **Date Generated**: 2026-06-02

📊 Performance & Reconstruction Quality Metrics

This table summarizes training time, VRAM consumption, Gaussian model size (number of Gaussians), and reconstruction quality metrics (PSNR, SSIM, LPIPS) across all training configurations and image scales.

Configuration	Image Scale	Initial Gaussians	Final Gaussians	Training Time	Peak VRAM (MiB)	PSNR (dB)	SSIM	LPIPS
FastGS Base	`images` (1x)	155,767	208,238	10m 01s	5,562.99	29.17	0.9069	0.2036
FastGS Base	`images_2` (2x)	155,767	211,181	9m 25s	5,194.35	29.03	0.8984	0.2218
FastGS Base	`images_4` (4x)	155,767	169,648	3m 54s	1,533.18	29.28	0.9084	0.1340
FastGS Base	`images_8` (8x)	155,767	129,718	2m 22s	582.47	29.73	0.9246	0.0803
FastGS Big	`images` (1x)	155,767	468,793	14m 04s	6,177.02	29.56	0.9170	0.1772
FastGS Big	`images_2` (2x)	155,767	478,529	13m 21s	5,814.90	29.45	0.9088	0.1951
FastGS Big	`images_4` (4x)	155,767	291,170	5m 31s	1,837.48	29.67	0.9168	0.1154
FastGS Big	`images_8` (8x)	155,767	164,747	2m 47s	693.32	30.00	0.9289	0.0702

🔍 Key Insights & Analysis

1. Resolution Downsampling Metrics Artifact

Mathematically, metrics such as PSNR and SSIM appear higher (and LPIPS appears lower/better) at lower resolutions like `images_8` and `images_4` compared to `images`. This occurs because downsampling acts as a low-pass filter, blurring away fine textures, high-frequency details, and sharp edges. Consequently, the network has less complex information to match, leading to higher mathematical scores. However, the absolute visual fidelity and detail representation are highest when trained on the full resolution `images` folder.

2. Base vs. Big Configuration Trade-off

- **Base Mode**: Optimizes for training speed and low memory footprint. A standard run on `images_4` completes in **3m 54s** using only **1.5 GB VRAM**.
- **Big Mode**: Prioritizes reconstruction accuracy by densifying 5x more frequently (`--densification_interval 100` vs `500` in Base) and utilizing a lower absolute gradient split threshold (`--grad_abs_thresh 0.0004` vs `0.0008` in Base). This produces a denser representation (e.g., **468k** vs **208k** Gaussians on `images`), yielding a significant bump in rendering fidelity (+0.39 dB PSNR and lower LPIPS on the full resolution).

💡 Recommendation for Maximum Quality Output

To generate the highest possible fidelity output for any scene in FastGS:

1. Use the **Big Configuration** (`run_fastgs_big.sh`).
2. Train on the full-resolution `images` folder.
3. If your source images exceed 1.6K pixels in width, they are downsampled to 1600px by default. Force training on the raw resolution by adding the `-r 1` flag to the training command (e.g. `python train.py -r 1 ...`).